9.6 Hydrated salts and Water of Crystallization

When ionic substances are dissolved in water to make their saturated aqueous solution, their ions attract water molecules which then attached chemically in certain ratio. This process is called hydration. These ionic substances crystallize out from their saturated aqueous solution with a definite number of molecules of water. The number of water molecules found in the crystalline substance is called **water of crystallization**. Such salts are called hydrated salts.



Figure 9.8 a) Crystalline hydrated salt
b) Amorphous anhydrous salt

On heating these hydrated crystalline salts, they lose their water of crystallization and become amorphous or lose their colour (if they are coloured). Table 9.3 shows some common hydrated salts:

Table 9.3 Hydrated salts

Common Name	IUPAC Name	Molecular Formula
Blue Vitriol	Copper (II) sulphate pentahydrate	$CuSO_4 \cdot 5H_2O$
Epsom Salt	Magnesium sulphate heptahydrate	$MgSO_4 \cdot 7H_2O$
Gypsum	Calcium sulphate dihydrate	$CaSO_4 \cdot 2H_2O$
Green Vitriol	Iron (II) sulphate heptahydrate	$FeSO_4 \cdot 7H_2O$
White Vitriol	Zinc sulphate heptahydrate	$ZnSO_4 \cdot 7H_2O$

9.6.1 Copper sulphate pentahydrate $CuSO_4 \cdot 5H_2O$ (Blue vitriol)

The number of water molecules in blue vitriol is five. So its water of crystallization is 5. When blue coloured copper sulphate crystals are gently heated, it loses its five water molecules and becomes colourless anhydrous copper sulphate.

$$\underset{\text{(Copper sulphate pentahydrate)}}{CuSO_4 \cdot 5H_2O} \underset{\text{Cooling}}{\overset{\text{Heating}}{\rightleftharpoons}} \underset{\text{(Anhydrous copper sulphate)}}{CuSO_4 + 5H_2O}$$

Blue colour — colourless

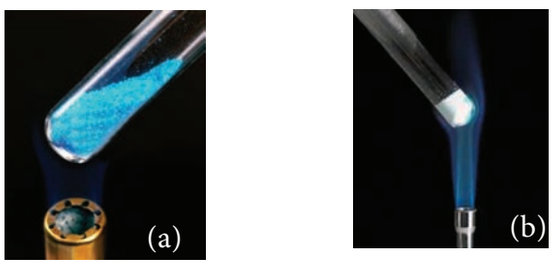


Figure 9.9 a) Copper sulphate before heating
b) Copper sulphate after heating

If you add few drops of water or allow it to cool, the colourless anhydrous salt again turns back into blue coloured hydrated salt.



Figure 9.10 Anhydrous copper sulphate turns to blue when water is added

9.6.2 Magnesium sulphate heptahydrate $MgSO_4 \cdot 7H_2O$ (Epsom salt)

Its water of crystallization is 7. When magnesium sulphate heptahydrate crystals are

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